

Assignment 2: Take a sample dataset and use Apriori algorithm to generate frequent itemset and rules of interest for the given support and confidence. (Take appropriate threshold for support and confidence).

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# Step 1: Import libraries
import pandas as pd
from mlxtend.frequent_patterns import apriori, association_rules

# Step 2: Load dataset
# Replace PLACEHOLDER_FILE_PATH with the path of your dataset file

data = pd.read_csv("/content/Groceries_dataset.csv")

# Step 3: View dataset
print("Dataset Preview:")
print(data.head())

# Step 4: Convert dataset into basket format
# Replace PLACEHOLDER_TRANSACTION_COLUMN with the column representing transaction id
# Replace PLACEHOLDER_ITEM_COLUMN with the column representing items

basket = (data
          .groupby(['Member_number', 'itemDescription'])
          ['itemDescription']
          .count()
          .unstack()
          .fillna(0))

# Step 5: Convert values to 0/1 (one-hot encoding)

basket = basket.applymap(lambda x: 1 if x > 0 else 0)

print("\nBasket Format:")
print(basket.head())

# Step 6: Find frequent itemsets using Apriori
# Replace PLACEHOLDER_SUPPORT with a support value like 0.02 or 0.01

frequent_itemsets = apriori(
    basket,
    min_support=0.05,
    use_colnames=True
)

print("\nFrequent Itemsets:")
print(frequent_itemsets)
```

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# Step 7: Generate association rules
# Replace PLACEHOLDER_CONFIDENCE with a value like 0.5 or 0.6

rules = association_rules(
    frequent_itemsets,
    metric="confidence",
    min_threshold=0.5
)

# Step 8: Show important metrics
print("\nAssociation Rules:")
print(rules[['antecedents', 'consequents', 'support', 'confidence', 'lift']])

# Step 9: Save results to CSV (optional)

rules.to_csv("association_rules_output.csv", index=False)

print("\nAssociation rules saved to association_rules_output.csv")
```

[illegible]

